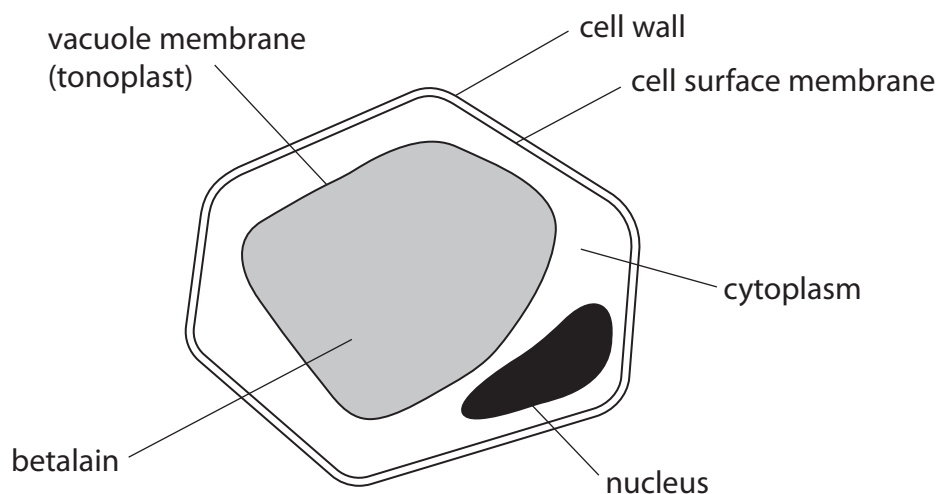


- 2 Scientists investigated the effect of salts on the permeability of cell membranes of beetroot.

The diagram shows a beetroot cell with a vacuole containing the red pigment betalain.



Equal sized discs of tissue were cut from a beetroot.

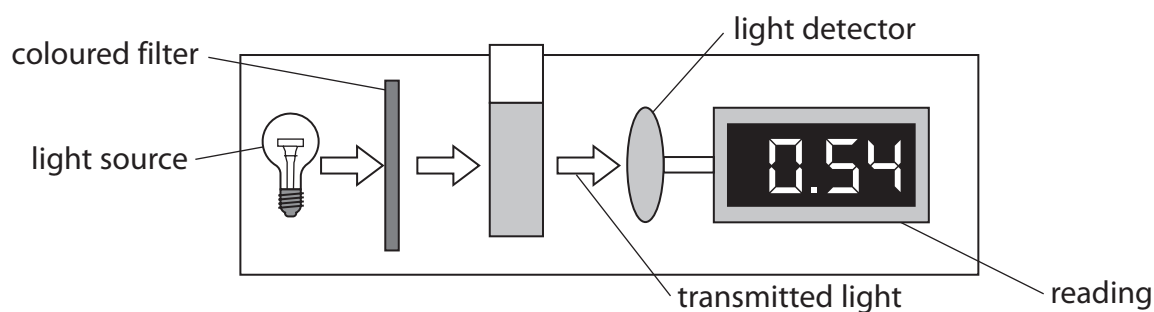
Twenty of these discs were placed into a beaker containing a calcium chloride solution.

This was repeated for several concentrations of calcium chloride and several concentrations of ammonium sulfate.

Betalain leaked from the discs, changing the colour of the solutions in the beakers. The intensity of each colour was recorded after 12 hours.



- (a) The diagram shows a colorimeter, which is an instrument that can measure the amount of light transmitted through a coloured solution.



The transmission of light through the solution decreases as the intensity of the colour increases.

- (i) Explain how using a colorimeter allows valid measurements of light transmitted through the betalain solutions.

(3)

- (ii) Temperature and pH affect the rate of leakage of betalain from discs of beetroot.

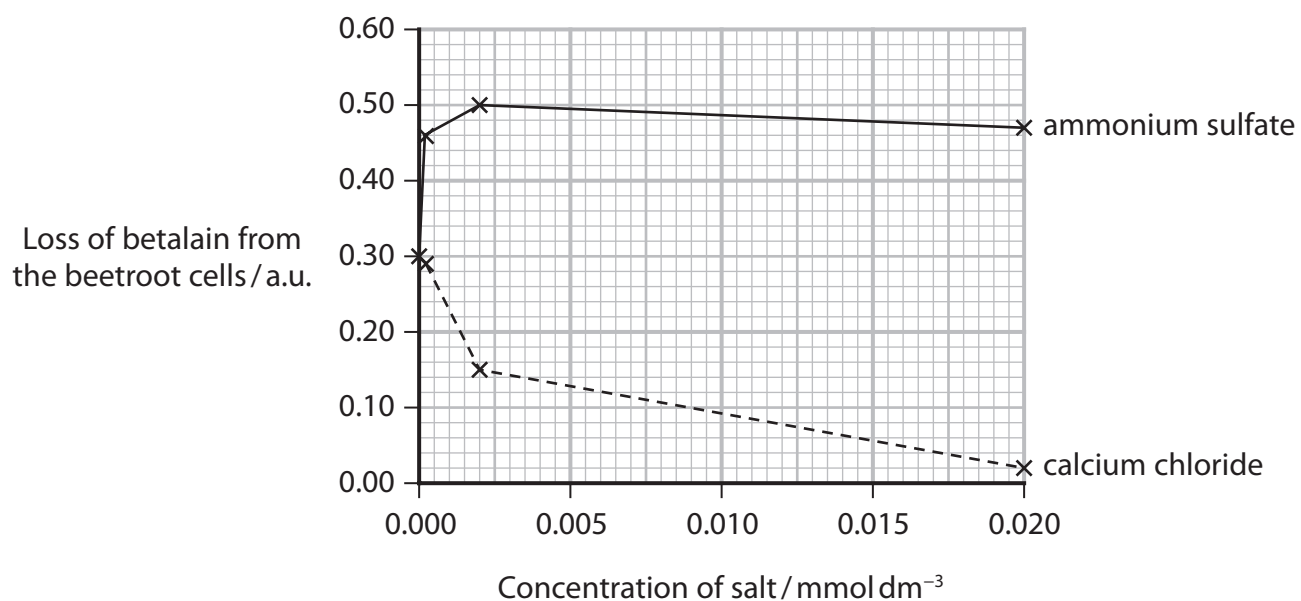
Describe how each of these variables could be kept constant in this investigation.

(2)

Temperature

pH

(b) The results of this investigation are shown in the graph.



(i) Draw a table to show the results of this study.

(3)



- (ii) Compare and contrast the effect of these two salts on the loss of betalain from the beetroot cells.

(3)

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- (iii) Calculate the percentage difference in colour intensity between calcium chloride and ammonium sulfate at a concentration of $0.01 \text{ mmol dm}^{-3}$.

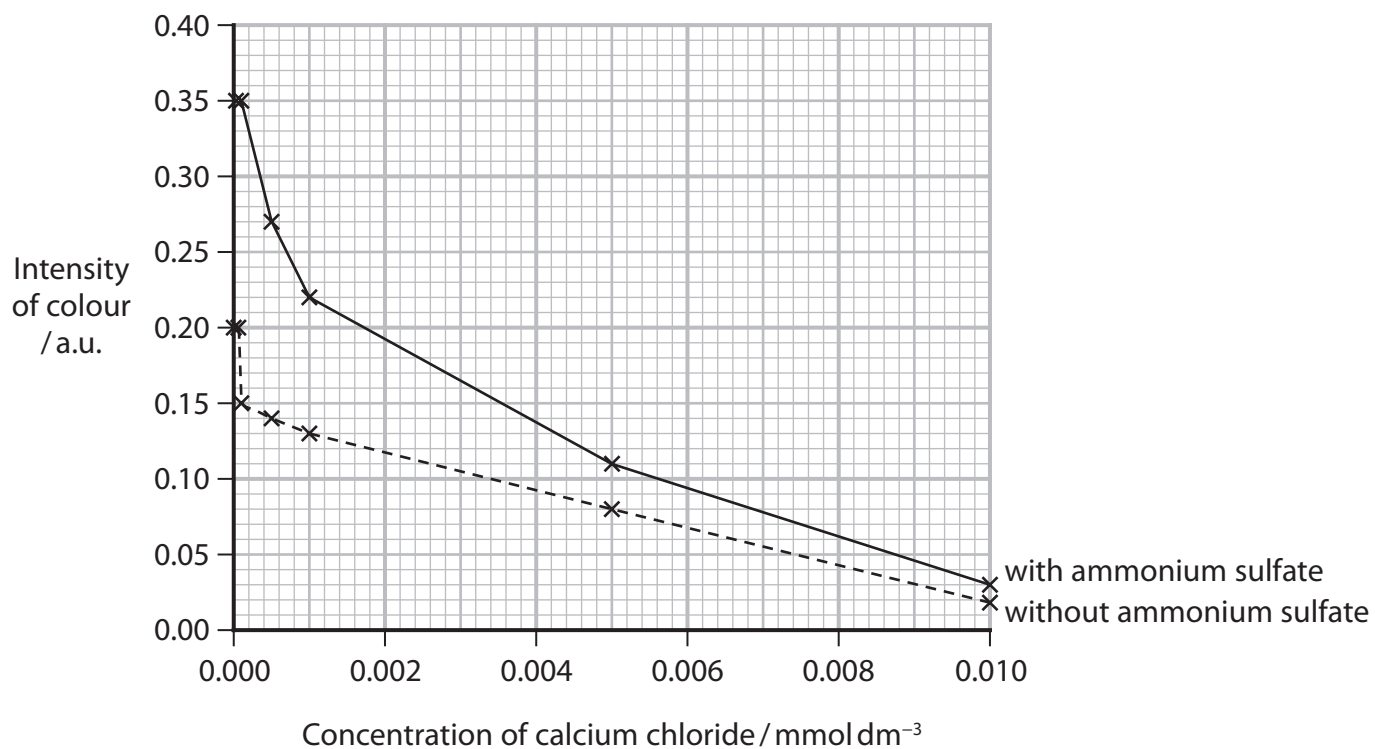
(2)

Answer %



- (c) In a second investigation, the effect of calcium chloride in the presence and in the absence of ammonium sulfate was studied.

The results are shown in the graph.



- (i) Comment on the results shown in the graphs from these two investigations.

(3)

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- (ii) Explain how you could modify the method to decide if the difference in colour intensity, at a concentration of $0.002 \text{ mmol dm}^{-3}$, is statistically significant.

(3)

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(Total for Question 2 = 19 marks)

